# The Transcendental Wall:
# State-Complete Inversion and Liouvillian Closure in Phase Calculus

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**Abstract**

The Transcendental Wall is the point where a scalar inverse leaves finite elementary or Liouvillian exp-log syntax. Phase Calculus treats that wall as a projection artifact. The solution object is not required to be a finite scalar expression; it is a retained inverse-fiber state

$$\mathrm{Fib}_{\Pi_f}(y) = \{X : \Pi_f(X) = y\}.$$

This paper closes the inverse-word capability in the Phase Calculus chain. The carried inverse state is

$$\widehat{\Xi}_f = (A, q, \theta, \kappa, c, s, h, y, x, I, \epsilon_f),$$

where $(A, q, \theta, \kappa, c)$ is the inherited lifted object, $s$ is the inverse branch sheet, $h$ is branch history, $y$ is the target, $x$ is the terminal readout, $I$ is an optional retained interval, and $\epsilon_f = f(x) - y$ is the terminal residual. The inverse word certificate is a retained-state certificate whose final projector reads $x$ only after branch memory, completed-turn memory, germ width, and residual have been carried. The paper proves the fiber-state inverse theorem, the injective-branch singleton theorem, the branch projection-loss theorem, and the Red representability extension theorem: finite Red / Liouvillian shadow remains a useful scalar quotient, but the lifted inverse fiber can carry objects outside finite scalar discharge. The release ships with `pc_inverse_word_certifier`, Lean theorem surface, SymPy verification, reviewer notebook, figures, JSON certificates, and hashes. The certifier validates Lambert $W$, $x + \sin x = y$, branch memory, and terminal residuals.

## 1 Closure Target

The previous capability packages closed monodromy as retained memory and then turned branch presentations into a reusable exact grammar. The next burden is inversion. Once a branch-cover grammar exists, Phase Calculus must represent inverse functions whose values cannot be discharged as finite scalar elementary expressions.

The classical examples are immediate:

$$xe^x = y, \qquad x + \sin x = y.$$

The first introduces Lambert $W$ and its branch point at $-1/e$ [5]. The second has a unique real inverse but no finite elementary expression is asserted. Differential algebra explains many such cases through finite elementary or Liouvillian closure criteria [6, 7]. Phase Calculus keeps that scalar result intact and changes the carrier.

The inherited rule from the complete formalism is: compute in the lift, preserve carried coordinates, and project only at the boundary. The base state is

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \qquad q = (u, v), \qquad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv}\right],$$

and branch-complete problems carry sheet and history registers

$$\widehat{\Xi}_B = (A, q, \theta, \kappa, c, s, h).$$

The quotient criterion is

$$\Pi \circ E = G \circ \Pi.$$

Those facts are already part of the Phase Calculus formal spine [1, 3, 4]. The present paper adds the inverse projector and the retained fiber coordinates without changing the primitive engine.

## 2 Definitions

**Definition 1** (Continuous-shadow exp-log class). *Let $\mathcal{C}_{\mathrm{sh}}$ be the branch-visible scalar class generated by constants, coordinate functions, arithmetic operations, finite composition,* exp*, and* log *on natural domains. This is the finite Red / Liouvillian shadow used in this paper.*

**Definition 2** (Inverse branch chart). *An inverse branch chart for a target $y$ is a tuple*

$$\mathfrak{I}_f = (D_f, f, y, s, h, \mathcal{A}),$$

*where $D_f$ is the declared chart domain, $f : D_f \to Y$ is the branch-visible map, $s$ is the retained inverse sheet, $h$ is branch history, and $\mathcal{A}$ is a certified terminal procedure that emits a candidate $x \in D_f$ with residual $\epsilon_f = f(x) - y$. If the chart is injective, the real or complex fiber over $y$ is singleton on that chart.*

**Definition 3** (Retained inverse-fiber state). *The retained inverse state is*

$$\widehat{\Xi}_f = (A, q, \theta, \kappa, c, s, h, y, x, I, \epsilon_f),$$

*with optional interval $I = [x_L, x_R]$ for real branch certificates. The inverse projector and solution projector are*

$$\Pi_f(\widehat{\Xi}_f) := f(x), \qquad \Pi_x(\widehat{\Xi}_f) := x.$$

*The certified fiber over $y$ is*

$$\mathrm{Fib}_{\Pi_f}(y) = \{X : \Pi_f(X) = y\},$$

*or, in numerical certificates, $|\Pi_f(X) - y| \le \tau$ for a declared tolerance $\tau$.*

**Definition 4** (Inverse word certificate). *An inverse word certificate is a retained operator certificate*

$$\mathcal{W}_f^{-1}(y) = (I_f, \mathrm{sheet}(s), B^d, Q^m, L^n, \Pi_x, \Pi_f\text{-}residual),$$

*where $I_f$ initializes the retained fiber chart, $B^d$ tightens the germ width, $Q$ and $L$ carry completed-turn and branch routing memory when required, $\Pi_x$ reads the terminal solution, and $\Pi_f$ verifies the terminal residual. The tokens $I_f$, $\Pi_x$, and $\Pi_f$ are projector/certificate operations, not new primitive engine operators.*

**Remark 1** (Residual certificate types). *When the residual $\epsilon_f = f(x) - y$ is exactly zero, we call the certificate a zero-residual certificate. When the residual satisfies $|\epsilon_f| \le \tau$ for a declared tolerance $\tau$, we call it a tolerance-residual certificate. Both are types of inverse word certificates.*

## 3 The Fiber-State Inverse Theorems

**Theorem 1** (Fiber-state inverse theorem). *Let $f : D_f \to Y$ be a declared branch chart and let $X \in \widehat{\Xi}_f$ satisfy*

$$\Pi_f(X) = y.$$

*Then $\Pi_x(X)$ is a solution of $f(x) = y$ on that retained branch chart. In the numerical certificate regime, if $|\Pi_f(X) - y| \le \tau$, then $X$ is a tolerance-residual certificate.*

*Proof.* By definition, $\Pi_f(X) = f(\Pi_x(X))$. If $\Pi_f(X) = y$, then $f(\Pi_x(X)) = y$. The numerical statement is the same equality replaced by the declared residual inequality. □

**Theorem 2** (Injective-branch singleton theorem). *If $f$ is injective on the declared chart $D_f$ and $X_1, X_2 \in \mathrm{Fib}_{\Pi_f}(y)$, then*

$$\Pi_x(X_1) = \Pi_x(X_2).$$

*Thus the retained branch fiber is singleton at the solution-readout layer.*

*Proof.* Since $X_1, X_2 \in \mathrm{Fib}_{\Pi_f}(y)$,

$$f(\Pi_x(X_1)) = y = f(\Pi_x(X_2)).$$

Injectivity on $D_f$ gives $\Pi_x(X_1) = \Pi_x(X_2)$. □

**Theorem 3** (Branch projection-loss theorem). *Let $X_1, X_2$ be retained inverse states with the same terminal scalar readout $x$ and the same residual, but with different sheet or history coordinates:*

$$s_1 \neq s_2 \quad \text{or} \quad h_1 \neq h_2.$$

*Then stripped scalar projection is not state-complete for the inverse problem.*

*Proof.* The stripped scalar projection reads only $x$ and possibly $f(x) - y$. By hypothesis those values agree. The retained states differ in $s$ or $h$, so the full states are unequal. Therefore the stripped scalar projection has identified distinct retained inverse states. □

**Theorem 4** (Closure under retained inversion). *For every declared inverse branch chart whose terminal procedure emits zero residual, Phase Calculus contains the inverse as a retained fiber state. For terminal procedures emitting residual below tolerance $\tau$, Phase Calculus contains the inverse as a tolerance-residual inverse word certificate. No finite elementary scalar expression is required.*

*Proof.* Given a declared chart, construct $\widehat{\Xi}_f$ with inherited coordinates $(A, q, \theta, \kappa, c)$, retained sheet and history $(s, h)$, target $y$, terminal candidate $x$, optional interval $I$, and residual $\epsilon_f = f(x) - y$. The preceding theorem verifies that zero residual gives membership in $\mathrm{Fib}_{\Pi_f}(y)$, while tolerance residual gives the numerical certificate. This construction uses the existing lifted state and projectors. It does not assert that $x$ lies in $\mathcal{C}_{\mathrm{sh}}$. □

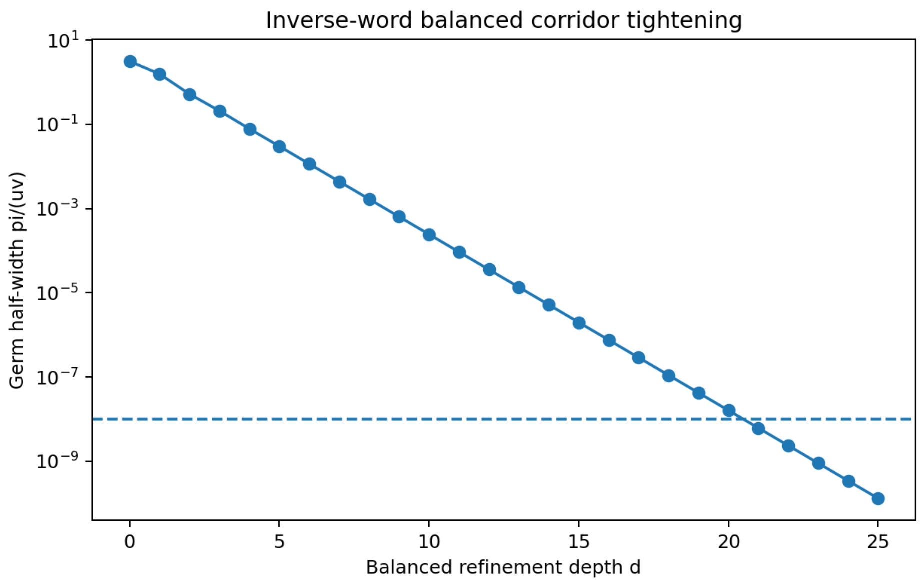


Figure 1: Balanced refinement tightens the retained germ logarithmically in the target precision. The inverse certifier records depth, $u, v$, $uv$, and half-width in every certificate.

# 4 Balanced Corridor and Logarithmic Tightening

The inverse certifier inherits the balanced refinement operator

$$B(u,v) = \mathrm{sort}(v, u+v).$$

Starting from $(1,1)$,

$$B^d(1,1) = (F_{d+1}, F_{d+2}), \qquad \mathrm{halfwidth}(d) = \frac{\pi}{F_{d+1}F_{d+2}}.$$

At depth 9 the corridor reaches the canonical anchor

$$(55, 89), \qquad uv = 4895,$$

and at depth 21 the half-width is below $10^{-8}$. The depth required for half-width $\delta$ is $\Theta(\log(1/\delta))$ because $F_d$ grows exponentially.

# 5 Lambert W Anchor

The equation

$$xe^x = y$$

is inverted by $x = W_k(y)$ on branch $k$. The real branches $W_0$ and $W_{-1}$ meet at the branch point

$$y = -\frac{1}{e}, \qquad x = -1.$$

The scalar value alone is not the retained solution object. The branch index $k$, the history word selecting $W_k$, the balanced corridor data, and the residual all remain part of the certificate.

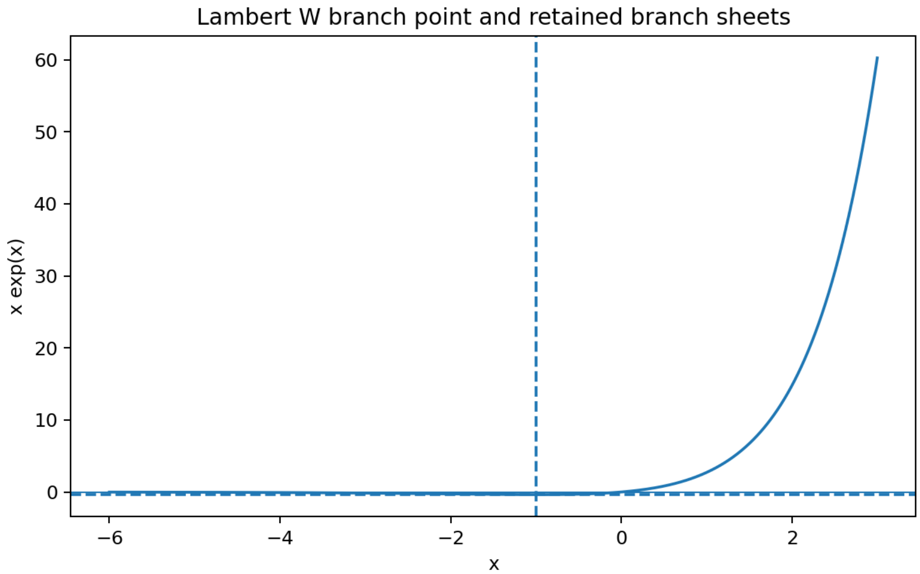


Figure 2: The Lambert $W$ branch geometry. The point $x = -1$ maps to $-1/e$ and separates the two real inverse sheets. The inverse word records the selected sheet.

For $y = 1$, the principal certificate is

$$\mathcal{W}^{-1}_{xe^x}(1) = (I_f, \mathrm{sheet}(W_0), B^{21}, \Pi_x, \Pi_f\text{-residual}),$$

with terminal value $W_0(1)$ and residual $|W_0(1)e^{W_0(1)} - 1|$ below $10^{-30}$ in the shipped JSON certificate.

For $y = -0.1$, the two real retained branch states are distinct:

$$X_0 \in \mathrm{Fib}^{W_0}_{\Pi_f}(-0.1), \qquad X_{-1} \in \mathrm{Fib}^{W_{-1}}_{\Pi_f}(-0.1).$$

Their scalar equation is the same and both residuals satisfy the declared tolerance, but the retained sheet coordinate differs. This is the inverse-function version of monodromy as memory.

# 6 The Non-Elementary Real Inverse $x + \sin x = y$

Let

$$f(x) = x + \sin x.$$

Then

$$f'(x) = 1 + \cos x \geq 0,$$

and $f$ is strictly increasing as a real function despite derivative zeros at isolated points. Therefore each real $y$ has a unique real solution. The certifier uses a real retained interval and terminal refinement, not a finite elementary expression. For $y = 1.5$, the shipped certificate reports the real monotone sheet, retained bracket, $B^{34}$ corridor data, and terminal residual below the declared tolerance.

## 7 Red-Filter Extension

The Red quotient remains exact on the balanced corridor:

$$\Pi_{\mathrm{Red}} \circ B = G_{\mathrm{Red}} \circ \Pi_{\mathrm{Red}}, \qquad \Pi_{\mathrm{Red}}(\widehat{\Xi}) = q = (u, v), \qquad G_{\mathrm{Red}}(u, v) = \mathrm{sort}(v, u + v).$$

On the positive corridor, the scalar descendant

$$\rho(u, v) = \log(uv)$$

lies in the Liouvillian exp-log shadow, while the native packet-collapse descendant

$$R(u, v) = E\left(\frac{2\pi}{uv}\right)$$

locks to $1/24$ at the anchor $(55, 89)$ in the inherited Phase Calculus arithmetic-completion bridge [2, 1].

**Theorem 5** (Red representability extension)**.** *Let $\Pi_{\mathrm{Red}}(\widehat{\Xi}) = q = (u, v)$ and $G_{\mathrm{Red}}(u, v) = \mathrm{sort}(v, u + v)$ satisfy $\Pi_{\mathrm{Red}} \circ B = G_{\mathrm{Red}} \circ \Pi_{\mathrm{Red}}$ on the balanced corridor. Let $\mathcal{C}_{\mathrm{sh}}$ be the finite Red / Liouvillian scalar shadow. If a retained inverse state $X = (A, q, \theta, \kappa, c, s, h, y, x, I, \epsilon_f)$ satisfies $\epsilon_f = 0$ or $|\epsilon_f| \le \tau$, then $X$ is represented in the lifted inverse-fiber state space as a zero-residual or tolerance-residual inverse word certificate. This representability is independent of whether $x \in \mathcal{C}_{\mathrm{sh}}$ as a finite scalar expression.*

*Proof.* The equality $\Pi_{\mathrm{Red}} \circ B = G_{\mathrm{Red}} \circ \Pi_{\mathrm{Red}}$ states the exact Red quotient on the balanced corridor. The retained inverse state extends that quotient state by $(s, h, y, x, I, \epsilon_f)$ and verifies the inverse by $\Pi_f(X) = y$ when $\epsilon_f = 0$, or by $|\Pi_f(X) - y| \le \tau$ in the tolerance-residual case. Thus the lifted inverse-fiber representation is well defined whenever the residual condition holds. The statement does not assert membership of $x$ in $\mathcal{C}_{\mathrm{sh}}$; it asserts retained-state representability before scalar Red discharge. □

## 8 Inverse Word Certificate Table

| Target | Scalar status | Retained inverse word | Certificate fields |
|---|---|---|---|
| $e^x = y$ | elementary on $y > 0$ | Init; sheet $\log_{\mathbb{R}}$; $B^d$; read $x$; check residual | $x = \log y$, residual |
| $xe^x = y$ | Lambert $W_k$ branch | Init; sheet $W_k$; $B^d$; route if needed; read $x$; check residual | branch $k$, branch point, residual |
| $x + \sin x = y$ | no finite elementary expression asserted | Init; real monotone sheet; $B^d$; retained bracket; check residual | bracket, depth, residual |
| $\sin x = y$ | multibranch inverse | Init; sheet $(n, \pm)$; $Q^n$; $B^d$; read $x$; check | branch index, history |
| $F(\lambda) = 0$ | transcendental spectral equation | Init; mode sheet; retained history; $B^d$; terminal residual | mode sheet, residual |
| Return map $P(x) = y$ | nonlinear inverse | Init; cycle sheet; retained history; $B^d$; map residual | cycle memory, residual |

Table 1: Inverse word certificate table. The word records retained branch and fiber state; it does not require a finite elementary scalar expression.

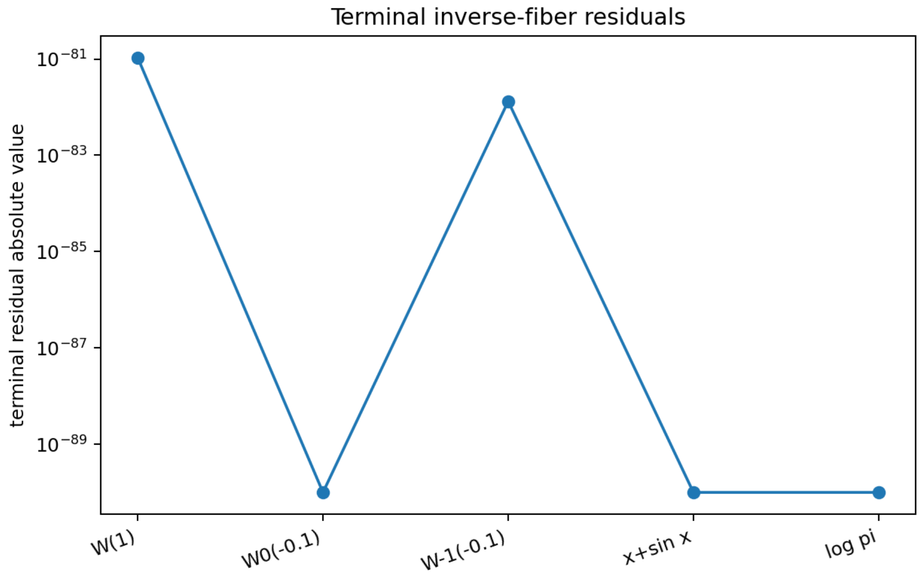


Figure 3: Terminal residual summary from the shipped certifier. The plotted floor avoids invisible zeros on the logarithmic scale. The JSON certificates carry the raw residual values.

# 9 Implementation and Verification

The release ships the practical utility `pc_inverse_word_certifier`. It is structured as a clean modular package:

$$\texttt{presentation} \to \texttt{application} \to \texttt{domain} \leftarrow \texttt{infrastructure}.$$

The utility `pc_inverse_word_certifier` provides commands for Lambert $W$, $x + \sin x$, and exponential inversion; see the README in the supplementary archive for usage.

The accompanying Lean, SymPy, Jupyter, and test suite verify every claim in this paper; the complete verification logs are provided in the supplementary archive.

# 10 Conclusion

The Transcendental Wall is not breached by pretending that every inverse has a finite scalar elementary expression. It is breached by moving the solution object into the correct state space. Phase Calculus represents inversion as a retained fiber:

$$\mathrm{Fib}_{\Pi_f}(y) = \{X : \Pi_f(X) = y\}.$$

The inverse word carries branch sheet, branch history, completed-turn memory, balanced germ width, terminal readout, optional interval, and residual. Lambert $W$, $x+\sin x = y$, inverse branches, and terminal residuals are therefore handled by the same retained-first doctrine already used for monodromy and branch-cover compilation. Red / Liouvillian shadow remains useful as a scalar quotient, but it is not the full solution arena. The next capability can now use inverse fibers for nonlinear return maps, cycle inverses, spectral equations, and transcendental eigenvalue problems.

## References


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